

Clinical Findings

Signs of acute riboflavin deficiency include deep red erythema, epidermal necrolysis, and mucositis. The severity of symptoms correlates with the degree of deficiency.

Clinical manifestations of chronic riboflavin deficiency (ariboflavinosis) typically appear 3 to 5 months after starting an inadequate diet. Skin and mucous membrane changes are predominant. Angular stomatitis initially presents as small papules at the corners of the mouth, which enlarge, ulcerate, and progress to macerated fissures extending laterally—often bleeding. Cheilosis may develop, characterized by erythema, xerosis, and vertical fissuring of the lips.

Early glossitis is marked by prominent lingual papillae; as the condition progresses and papillae are lost, the tongue becomes smooth, swollen, and magenta-colored.

The dermatitis associated with riboflavin deficiency resembles seborrheic dermatitis and commonly affects the nasolabial folds, nostrils, nasal bridge, forehead, cheeks, and posterior auricular regions. Flexural areas of the limbs may also be involved. Dysebacia—plugging of sebaceous glands—can be observed around the nose.

Genital involvement occurs more frequently and severely in males, presenting as a red, confluent, crusty, or lichenified dermatitis of the scrotum that may extend to the inner thighs. In general, lesions worsen in areas subject to chafing or trauma. Infants often show dermatitis in the inguinal region, while older individuals may have more pronounced involvement in facial creases and wrinkles. Incontinent patients may develop lesions in the perianal and buttock areas.

Cutaneous findings are not aggravated by light exposure but may worsen with heavy physical activity. Ocular manifestations, particularly photophobia and conjunctivitis, are also common. The term **oculo-oro-genital syndrome** describes this constellation of ocular, oral, and genital symptoms.

Laboratory Testing

A normochromic, normocytic anemia may be present. Erythrocyte glutathione reductase activity can be measured as a screening test, but a therapeutic trial of riboflavin supplementation is often the most effective way to confirm the diagnosis.

Treatment

The recommended daily intake of riboflavin is 0.6 mg per 1000 kcal.

- **Infants and children:** 1 to 3 mg per day
- **Adults:** 10 to 20 mg per day

Vitamin B₃ (Niacin)

Etiology and Pathogenesis

Niacin (vitamin B₃) functions as a coenzyme and can be obtained from the diet or synthesized endogenously from the essential amino acid tryptophan. Dietary sources include whole grains, enriched breads, nuts, dairy products, liver, animal meats, mushrooms, and dried beans.

Dietary niacin is primarily present as nicotinamide adenine dinucleotide (NAD) and NADPH. These are hydrolyzed in the intestinal lumen to nicotinamide, which may be converted to nicotinic acid by gut bacteria or absorbed directly into the bloodstream. Both forms are transported to the liver, kidneys, and intestines, where they are reconverted to NAD and NADPH. These coenzymes serve as hydrogen donors and acceptors in oxidation-reduction reactions critical for carbohydrate, fatty acid, and protein metabolism.

Epidemiology

Niacin (vitamin B₃) deficiency causes pellagra. In 1735, Gaspar Casal described a group of impoverished peasants in northern Spain suffering from a distinctive

skin disorder he termed "mal de la rosa," referring to the reddish, glossy rash on the dorsum of the hands and feet.